

Author Gallery - Upgrade Week Plan

Developer 3 (Backend Engineer) Action Plan

TARGET ROLE: BACKEND APIS & SCHEMAS

1. Scope of Work & Responsibilities

- **Primary Focus:** Database Schemas (Bookshelf, Comments, Forums, Series), REST API Endpoints, Aggregations for Analytics, AI Assistant Backend Stubs.
- **Git Feature Branch:** feature/backend-api-schemas
- **Target Workplaces:** backend/models, backend/controllers, backend/routes, backend/server.js

2. Git & Collaboration Commands

```
# 1. Clone repository & enter backend directory
git clone https://github.com/ManishBhattPC/Author-Gallery.git && cd Author-Gallery/backend
# 2. Checkout main & create feature branch
git checkout main && git pull origin main
git checkout -b feature/backend-api-schemas
# 3. Commit & push endpoints
git add . && git commit -m "feat: created bookshelf and comment models & routes"
git push origin feature/backend-api-schemas
```

3. Detailed File Implementation Checklist

- **[NEW] backend/models/Bookshelf.js:** Mongoose schema mapping user to book lists (wantToRead, reading, completed, favorites).
- **[NEW] backend/models/ChapterComment.js:** Mongoose schema for bookId, chapterIndex, userId, content, and createdAt timestamp.
- **[NEW] backend/models/ForumThread.js:** Mongoose schema for user discussions, categories (General, Writing Advice, Book Club), and replies array.
- **[NEW] backend/models/BookSeries.js:** Mongoose schema for grouping books into volume/series collections.
- **[NEW] backend/controllers & routes/bookshelfRoutes.js:** GET /api/bookshelves, POST /api/bookshelves/add, DELETE /api/bookshelves/remove.
- **[NEW] backend/controllers & routes/commentRoutes.js:** GET /api/comments/:bookId/:chapterIndex, POST /api/comments/create.
- **[NEW] backend/controllers & routes/discussionRoutes.js:** GET /api/discussions, POST /api/discussions/create, POST /api/discussions/:id/reply.
- **[NEW] backend/controllers & routes/aiAssistantRoutes.js:** POST /api/ai-assistant/grammar-check, POST /api/ai-assistant/summarize, POST /api/ai-assistant/plot-ideas.
- **[MODIFY] backend/controllers/dashboardController.js:** Aggregate daily view stats, PDF download metrics, and rating breakdowns for charting.
- **[MODIFY] backend/server.js:** Register all 4 new API router middlewares under /api/.

4. Verification & Testing Steps

- Test all new endpoints with Postman or Thunder Client.
- Verify authMiddleware protection on write endpoints (Bookshelf, Comments, Discussions).
- Ensure MongoDB indexes exist for bookId and userId lookups.